



Applications of numerical series.

Laser cavity gain

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Abstract

This work explores the application of numerical series in the calculation of the optical gain of a resonant laser cavity. The operating principle of lasers, their basic physical structure, and the processes of spontaneous emission, absorption, and stimulated emission that govern their operation are described. An analytical expression for the threshold gain is developed from the Taylor series expansion of the exponential function, considering the reflection losses at the cavity mirrors. It is shown that the linear approximation $e^{-2\alpha L} \approx 1 - 2\alpha L$ is physically justified for the vast majority of high-reflectivity industrial lasers, and the truncation error is quantitatively analyzed. The obtained result, $\alpha_c = (1 - R_1 R_2)/(2L)$, is directly useful in the design and optimisation of high-power lasers for advanced metallurgical manufacturing processes, such as selective laser sintering (SLM), direct metal laser sintering (DMLS), and high-precision laser machining.

Keywords: LASER, resonant cavity, optical gain, Taylor series, threshold condition, laser manufacturing processes, metallurgical engineering.

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Introduction

The LASER (*Light Amplification by Stimulated Emission of Radiation*) is a device that operates on the principle of stimulated emission, used to generate a coherent light beam —both spatially and temporally— which allows producing highly concentrated, monochromatic light beams of great intensity.

LASER devices have numerous applications in modern life: from audio reproduction on optical media (CD/DVD) to nuclear energy applications in inertial confinement fusion laboratories. However, the main interest for metallurgical engineering lies in their application in advanced manufacturing processes:

- **Laser machining:** high-precision cutting, welding and drilling.
- **Metal 3D printing:** direct metal laser sintering (DMLS) and selective laser melting (SLM), core processes of industrial additive manufacturing.

This work focuses on the mathematical analysis of one of the critical components of the laser: the **optical resonant cavity**. The objective is to demonstrate how numerical series —specifically the Taylor series expansion— allow modelling the threshold condition and calculating the cavity gain. This parameter is fundamental to determine the minimum energy required for laser operation in industrial applications.

Components and Operating Principle

Basic components of a laser

A laser consists of three essential components (Figure 1):

1. **Optical resonant cavity:** a set of optical elements, usually two mirrors: one highly reflective ($\approx 100\%$) and one partially reflective ($\approx 95\%$) that allows the laser beam to exit.
2. **Active medium:** a substance —solid, liquid or gas— placed inside the resonant cavity where stimulated emission occurs.
3. **Pumping system:** the mechanism by which energy is supplied to the active medium; it can be optical, thermal, electrical or chemical.

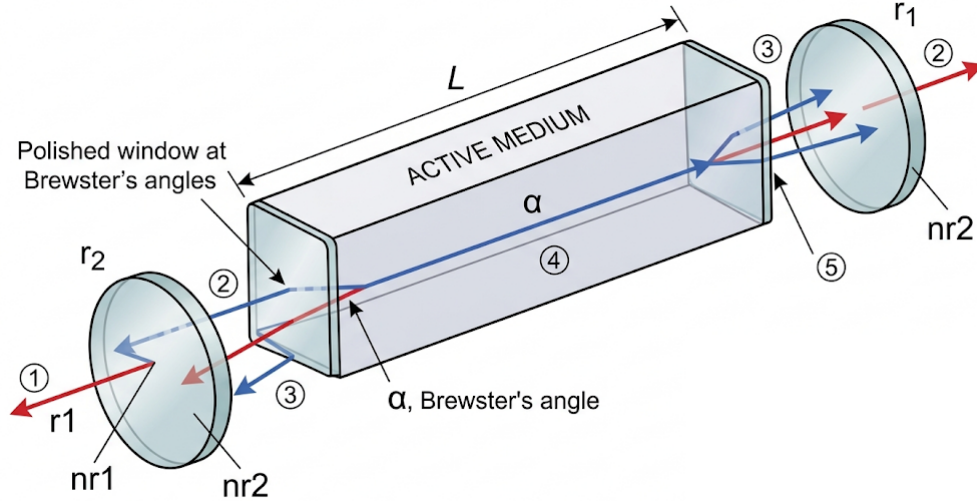


Figure 1: Schematic of a gas laser with Brewster windows, central active medium, high-reflectivity mirror (left) and semitransparent output mirror (right).

Operating principles — Quantum mechanics of the process

Electrons in atoms are found in discrete energy levels defined by quantum numbers. In their ground state, electrons neither absorb nor emit energy. The relation between the energy of a photon and the frequency of the emitted radiation is given by the Planck-Einstein relation:

$$\Delta E = h \nu \quad (1)$$

where ΔE is the photon energy, h is Planck's constant ($h = 6.626 \times 10^{-34}$ J·s) and ν is the radiation frequency.

Figure 2 illustrates the possible energy transitions between discrete quantum levels.

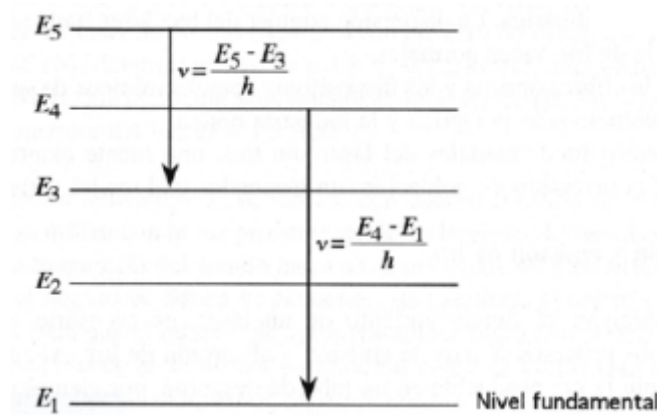


Figure 2: Energy level diagram. Arrows represent electronic transitions with photon emission at the corresponding frequencies.

When electrons absorb energy they jump to a higher level; when they decay to a lower level they emit photons. Normally, in the active medium, most atoms have their electrons in the ground state, but some excited atoms spontaneously emit photons

upon returning to that state. This process is random and the emitted photons are neither coherent nor maintain any phase relationship with each other.

When radiation of suitable energy impinges on an electron in an excited state, the atom is *stimulated* to emit a photon of the same frequency and phase. By adding both radiations —the incident and the emitted— **radiation amplification** occurs (Figure 3).

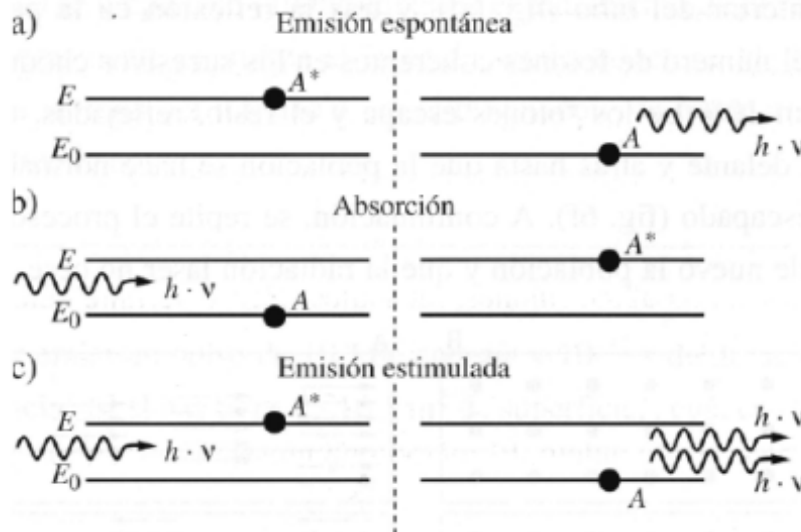


Figure 3: The three radiative processes: (a) spontaneous emission, (b) absorption and (c) stimulated emission. The latter is the one that sustains laser operation.

Under normal conditions, stimulated emission is a very small effect because few atoms are in an excited state. For a laser to work, the active medium must have at least three energy levels and the pumping system must achieve **population inversion**: bring most atoms to the excited state, making stimulated emission highly probable (Figure 4).

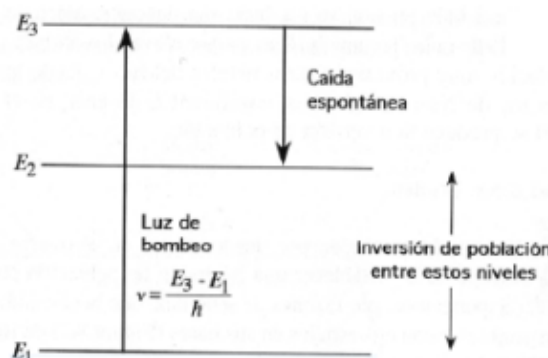


Figure 4: Population inversion in a three-level system. Pump light promotes electrons from E_1 to E_3 ; fast spontaneous decay to E_2 creates the inversion between E_2 and E_1 needed for stimulated emission.

The resonant cavity makes the light beams bounce back and forth inside it, gener-

ating phase-synchronised stimulated emission in pulses tuned to the cavity length L (Figure 5).

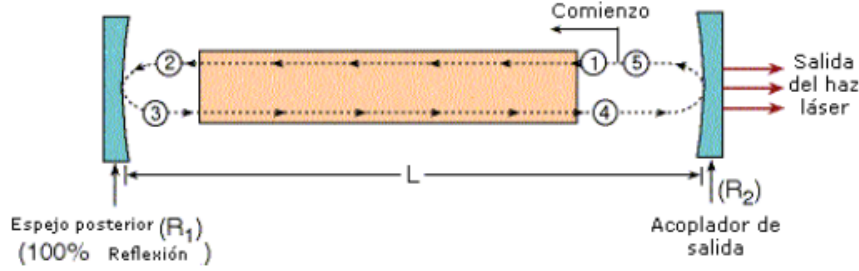


Figure 5: Fabry-Pérot resonant cavity. The beam travels round trips ($1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$) being amplified on each pass through the active medium. R_1 : high-reflectivity rear mirror; R_2 : semitransparent output coupler.

Cavity Threshold Condition

The **threshold condition** is the critical point at which the light gain inside the cavity exactly equals the losses, allowing the laser to begin sustained oscillation. To establish it, the following variables are defined:

- α [cm^{-1}]: gain per unit length in the active medium.
- L [cm]: length of the active material.
- R_1, R_2 (dimensionless, $0 < R_i \leq 1$): reflectivities of the two mirrors.
- ϕ [photons/ cm^3]: photon density in the cavity.

Consider a beam travelling parallel to the optical axis. On completing one *round trip* inside the cavity, the beam experiences:

1. **Gain:** by traversing the active medium twice (total distance $2L$), the photon density is multiplied by the amplification factor $e^{2\alpha L}$.
2. **Reflection losses:** by reflecting off both mirrors, the density is reduced by the factors R_1 and R_2 .

At threshold, the photon density at the beginning of the cycle ϕ_i must equal the density at the end ϕ_f :

$$\phi_i \cdot e^{2\alpha L} \cdot R_1 \cdot R_2 = \phi_f \quad (2)$$

Since $\phi_i = \phi_f$ in the steady-state regime at threshold, this simplifies to:

$$e^{2\alpha L} \cdot R_1 \cdot R_2 = 1 \implies e^{-2\alpha L} = R_1 \cdot R_2 \quad (3)$$

This is the **exact threshold equation**: a transcendental equation that relates the gain to the mirror reflectivities.

Mathematical Development Using Numerical Series

Taylor series expansion

At threshold, the gain α is a small quantity, so the product $2\alpha L \ll 1$. This allows us to approximate the exponential function in equation (3) using its Taylor series (Maclaurin series) around the origin. The general exponential series is:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad (4)$$

Substituting $x = -2\alpha L$ into the threshold equation (3):

$$e^{-2\alpha L} = \sum_{n=0}^{\infty} \frac{(-2\alpha L)^n}{n!} = 1 + (-2\alpha L) + \frac{(-2\alpha L)^2}{2!} + \frac{(-2\alpha L)^3}{3!} + \dots = R_1 R_2 \quad (5)$$

The series converges for all values of αL , and in particular converges absolutely for $2\alpha L < 1$, a condition that holds in high-reflectivity industrial lasers.

Truncation of the series — physical justification

Since α is small at threshold, the higher-order terms are much smaller than the first ones. Truncating series (5) after the linear term:

$$\boxed{e^{-2\alpha L} \approx 1 - 2\alpha L} \quad (6)$$

This approximation has a precise physical interpretation. Each pass through the cavity provides a small incremental gain; the higher-order terms of the series correspond to second-, third- and higher-order contributions of those accumulated corrections, which become negligible when $2\alpha L \ll 1$. Mathematically, the truncation corresponds to the first-order approximation $e^{-x} \approx 1 - x$ for $|x| \ll 1$, rigorously justified by the convergence criterion of the Taylor series.

Calculation of the Characteristic Gain

Substituting the approximation (6) into the threshold condition:

$$1 - 2\alpha_c L = R_1 R_2 \quad (7)$$

Solving for the **characteristic gain of the oscillator** α_c :

$$\boxed{\alpha_c = \frac{1 - R_1 R_2}{2L}} \quad (8)$$

This result is the central expression of this work. It is a simple and powerful analytic formula that directly yields the minimum threshold gain required for the laser to oscillate, given only the geometric and optical parameters of the cavity.

Truncation Error Analysis

The linear approximation (6) introduces a truncation error whose dominant term is the quadratic term of the series:

$$\varepsilon_{\text{trunc}} \approx \frac{(2\alpha L)^2}{2} \quad (9)$$

The *relative error* with respect to the linear term is of order αL :

$$\varepsilon_{\text{rel}} \approx \frac{(2\alpha L)^2/2}{2\alpha L} = \alpha L \quad (10)$$

High reflectivity case — typical industrial lasers

For a cutting laser with $R_1 = 0.99$, $R_2 = 0.95$ and $L = 10$ cm:

$$R_1 R_2 = 0.9405 \quad (11)$$

$$\alpha_c = \frac{1 - 0.9405}{2 \times 10} = \frac{0.0595}{20} = 2.975 \times 10^{-3} \text{ cm}^{-1} \quad (12)$$

$$\varepsilon_{\text{trunc}} = \frac{(2 \times 2.975 \times 10^{-3} \times 10)^2}{2} = \frac{(0.0595)^2}{2} \approx 1.77 \times 10^{-3} \quad (13)$$

The quadratic error is three orders of magnitude smaller than the linear term (0.0595). The approximation is excellent.

Low reflectivity or very long cavity case

If $R_1 R_2 = 0.5$ or L is very large, α grows and $2\alpha L$ ceases to be small. In this regime the linear approximation progressively loses validity. Table 1 illustrates the behaviour of the truncation error as a function of the product $2\alpha L$:

Table 1: Truncation error of the linear approximation as a function of $x = 2\alpha L$.

$x = 2\alpha L$	e^{-x} (exact)	$1 - x$ (approx.)	Relative error
0.05	0.9512	0.9500	0.13%
0.10	0.9048	0.9000	0.53%
0.20	0.8187	0.8000	2.28%
0.50	0.6065	0.5000	17.6%
1.00	0.3679	0.0000	100%

For values $x > 0.2$ it is advisable to include at least the quadratic term:

$$e^{-2\alpha L} \approx 1 - 2\alpha L + \frac{(2\alpha L)^2}{2} \implies \frac{(2\alpha_c L)^2}{2} - 2\alpha_c L + (1 - R_1 R_2) = 0 \quad (14)$$

which is a quadratic equation in α_c with an exact analytic solution, or alternatively solve the transcendental equation (3) numerically.

Discussion

Expression (8) is the central result of this work. Its derivation directly links **advanced calculus** —the theory of Taylor series— with a concrete design parameter in laser device engineering.

From an industrial design perspective:

- The formula allows a quick estimate of how much gain the active medium must provide for the laser to exceed threshold, without having to solve transcendental equations.
- The values of R_1 and R_2 are controllable design parameters (quality of mirror coatings), while L is a geometrical parameter of the cavity. The required gain α_c guides the choice of the active medium (doping density, amplification length).
- In SLM/DMLS processes, where high power density is required, short cavities with highly reflective mirrors are preferred, keeping $2\alpha L$ small and validating the linear approximation.

The convergence of the series and the validity of the linear approximation are justified by typical high-reflectivity operating conditions ($R_1 R_2 \gtrsim 0.9$, $x = 2\alpha L \lesssim 0.1$), where the truncation error is below 0.5%.

Conclusion

In this work, we have demonstrated the direct application of numerical series —specifically the Taylor series— to a fundamental problem in laser engineering: the calculation of resonant cavity gain.

Starting from the physical threshold condition for photon density, we arrived at the transcendental equation $e^{-2\alpha L} = R_1 R_2$. By expanding this exponential in a Maclaurin series and truncating to the linear term, we mathematically justified the approximation $e^{-2\alpha L} \approx 1 - 2\alpha L$, and obtained the analytical expression for the characteristic gain:

$$\alpha_c = \frac{1 - R_1 R_2}{2L}$$

The truncation error analysis confirms that the approximation is excellent (error $< 0.5\%$) for the typical operating conditions of industrial high-power lasers, where reflectivities are very high ($R_1 \approx 100\%$, $R_2 \approx 95\%$).

We conclude that advanced calculus, through the use of numerical series, provides essential tools to model and simplify complex problems in laser device physics. This modelling is fundamental to determine the minimum energy required to activate an industrial high-power laser, thus optimising cutting-edge metallurgical processes such as selective laser sintering and high-precision laser machining.

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